

TrueBench Methodology Overview

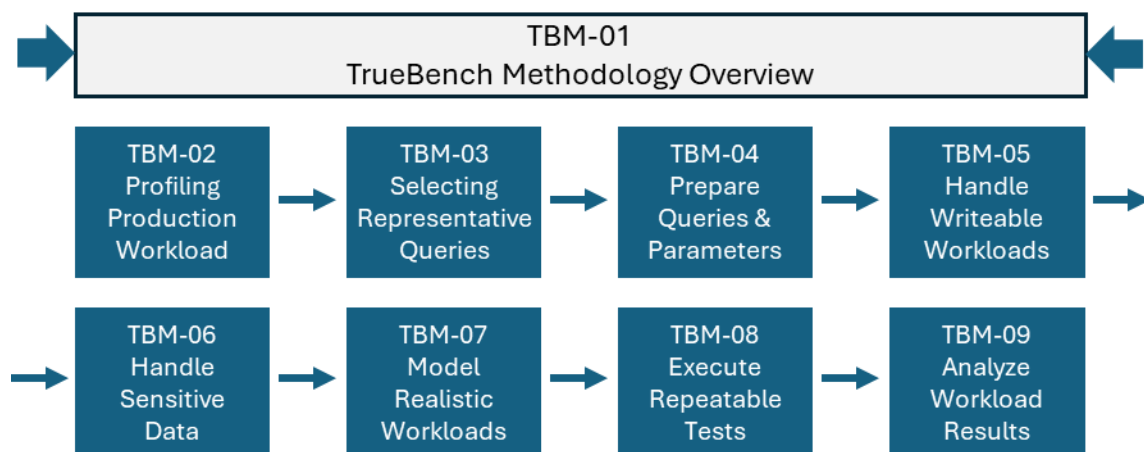
A practical framework for validating database and cloud Information systems using repeatable workload testing.

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Introduction

Modern information systems integrate dozens of components **from multiple vendors**. Databases, cloud infrastructure, query engines, storage systems, operating systems, networking layers, and analytic platforms all interact to deliver information to users.

These components evolve continuously. Software versions change, cloud infrastructure is updated, workloads shift, and new tools are introduced. The increasing use of AI-driven development is accelerating the rate of change across the entire technology stack.

Even when individual components appear to function correctly, changes in one part of the system can affect performance elsewhere in unexpected ways.

As a result, organizations increasingly face a fundamental question:

How can we verify that a complex information system continues to perform as expected as the environment evolves?

Application Changes

Schema Changes

Cloud Platform
Changes

Network Changes

Open-Source Library
Changes

Limitations of Traditional Monitoring

Monitoring tools play an important role in production operations. They measure resource utilization, detect outages, and provide alerts when system metrics exceed expected ranges.

However, monitoring systems typically focus on **aggregate statistics**, such as:

- CPU utilization
- query counts
- response time averages
- resource consumption summaries

Production Monitoring
(Platform)



System Metrics
SLA Tracking



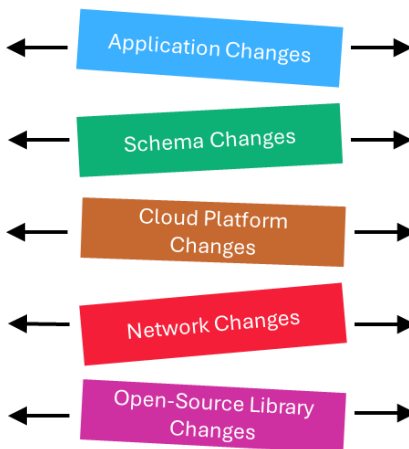
SLA Status
(Met / Missed / Alert)

These measurements are valuable for detecting system instability or capacity problems, but they do not always reveal subtle changes in workload behavior.

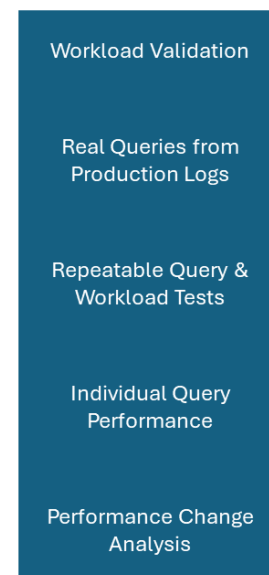
Performance changes often begin as **small degradations affecting specific queries or user activities**. These changes may not significantly alter overall system metrics, but can still impact individual users' experience.

In complex environments, these degradations may remain hidden until users begin reporting performance problems.

Traditional Monitoring



TrueBench Validation



TrueBench Validation

TrueBench validation takes a different approach to understanding system performance.

Instead of passively observing production systems, workload validation **actively executes controlled workloads** that represent real user activity.

By executing the same workload repeatedly under controlled conditions, it becomes possible to observe how system behavior changes over time.

This approach allows engineers to detect performance changes caused by:

- software upgrades
- infrastructure changes

- workload evolution
- configuration changes
- platform migrations

Workload validation provides **evidence-based insight** into system behavior rather than relying solely on passive monitoring metrics.

Principles of Effective Workload Validation

Meaningful workload validation requires several key principles.

Representative Workloads

Test workloads should reflect real production behavior rather than synthetic or simplified query sets.

Repeatability

Tests must be repeatable so that results from different test runs can be compared accurately.

Isolation

Tests should isolate the behavior of the system components being evaluated rather than introducing unrelated variables.

Observability

Test execution must capture sufficient metadata and performance information to allow analysis of the results.

TrueBench

TrueBench is a lightweight workload testing framework designed to support repeatable workload validation across modern information systems.

The framework allows engineers to:

- define test workloads using simple scripts
- execute queries with controlled concurrency

- support parameterized queries to simulate realistic workload variation
- integrate operating system and ETL activities within workload tests
- automate pre-test scripts to provision or prepare test data
- capture test metadata automatically, documenting individual query performance
- maintain linkage to host DBMS query logging to explain individual query behavior
- analyze performance differences between test runs

TrueBench evolved from TdBench, a benchmarking tool refined over more than fifteen years of real-world performance assessments.

The new Python-based implementation simplifies installation and test definition while extending the approach to modern database and cloud environments.

The TrueBench Methodology

The TrueBench methodology converts observed production workload behavior into repeatable workload tests that can be executed under controlled conditions to validate system performance.

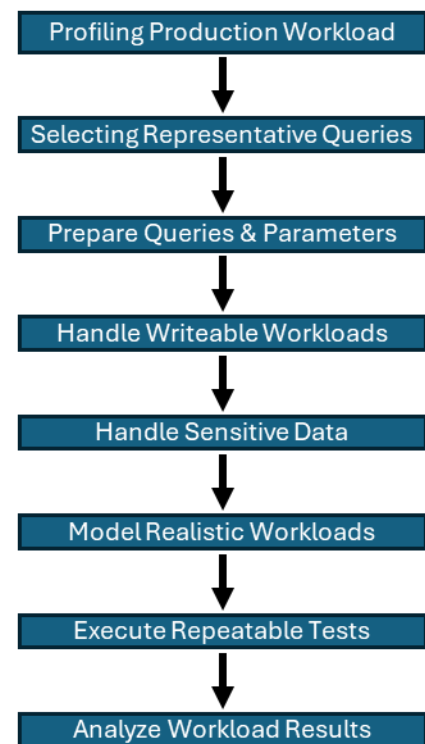
The methodology includes the following stages:

1. Profile Production Workloads

The process begins by examining query logs and system activity to understand how the system is actually used. Profiling reveals the tables most frequently accessed, the types of queries executed, and the concurrency patterns that define real production behavior.

2. Select Representative Queries

From the profiled workload, a representative set of queries is selected to reflect the diversity of production activity. The objective is not to reproduce every query, but to capture a mix that represents the overall workload characteristics.



3. Prepare Queries and Parameters

Queries extracted from logs often contain hard-coded values that reflect conditions at the time they were executed. These constraints must be converted into parameterized queries, and parameter values extracted from current data so the queries remain relevant during testing.

4. Model Realistic Workloads

Once queries are prepared, they are organized into workload models that reflect production behavior. These models define concurrency levels, workload mix, and execution pacing so that the test workload mirrors real system demand.

5. Handle Writable Workloads

Many real workloads include update activity, such as ETL processes or transactional modifications. Workload tests must simulate this behavior while protecting production data and ensuring that modified tables can be reset between test runs.

6. Handle Sensitive Data

In environments where production data cannot be used directly due to privacy or security restrictions, alternative datasets must be created that preserve the structural and statistical characteristics of the original data.

7. Execute Repeatable Tests

Workload tests are executed under controlled conditions so that results from different runs can be compared. TrueBench automates workload execution while capturing detailed metadata about query performance and system behavior.

8. Analyze Workload Results

The final step analyzes performance behavior across test runs to identify changes in query performance, system scalability, or workload balance. TrueBench links test results with database query logging to help explain observed performance differences.

These steps form a structured process for converting production workload behavior into repeatable workload validation tests.

The documents in the **TrueBench Methodology Series** describe each stage in detail.

Conclusion

Modern information systems evolve continuously. Software updates, infrastructure changes, and shifting workloads can influence performance in ways that are difficult to detect through monitoring alone. AI is accelerating the rate of change.

Workload validation provides a practical method for detecting these changes by executing controlled, repeatable workloads that represent real user activity.

The TrueBench methodology provides a structured framework for implementing this approach, enabling engineers to observe performance behavior and identify changes before they impact production users.

The documents in the TrueBench Methodology Series describe the individual steps required to build and execute realistic workload validation tests.